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B.Tech Mid-Term Examinations – October 2025

Third Semester

Artificial Intelligence & Data Science

# 23MAT204 MATHEMATICS FOR INTELLIGENT SYSTEMS III

Duration: Two hours | Maximum: 50 Marks

**Course Outcomes (COs):**

| CO | Course Outcomes |
|---|---|
| CO01 | Apply the fundamental techniques of optimization theory to solve data science problems. |
| CO02 | Analyse and solve computationally, physical systems using the formalism of partial differential equations. |
| CO03 | Apply Markovian concepts in stochastic sequential systems. |
| CO04 | Explain Bells Inequality and Quantum gates. |

**Answer all questions**

**Group A: Descriptive Type Questions**

1) Use the method of Lagrange multipliers to find the dimensions of a rectangle with the largest area, subject to the condition that the perimeter is at most 80 units.

[5] [CO01] [BTL 4]

2) A factory manufactures two products: A and B.
   - Each unit of A requires 5 hours of machine time and 14 hours of labor.
   - Each unit of B requires 5 hours of machine time and 7 hours of labor.
   - The factory has at most 15 hours of machine time and 35 hours of labor available per week.
   - The profit is: Rs.20 per unit of A, and Rs.30 per unit of B.

   **Tasks:**

   (i) Formulate this situation as a Linear Programming (LP) Problem: define decision variables, write the objective function, and list the constraints.

   [3] [CO01] [BTL 6]

   (ii) Solve the LP Problem to find the optimal production quantities of A and B that maximize weekly profit, and compute the maximum profit.

   [**Hint:** Use graphical method]

   [2] [CO01] [BTL 3]

3) Linearity and Convexity:

   (i) Check Convexity of a Set Determine whether the following set S is convex. (Justify your answer) $S = \{(x, y) : x^2 + y^2 \leq 9\}$

(ii) Check Linearity of Functions Determine whether the following functions are linear. Justify your answer using the properties of linearity (additivity and homogeneity).

a) $f_1(x,y) = 3x - 2y + 5$

b) $f_2(x,y) = 2x + y$

[2+3] [CO01] [BTL 2]

4) (i) Find the differential equation of all spheres of fixed radius having their centers in x y- plane.
(ii) Find the partial Differential Equation by eliminating arbitrary functions from $z = f(x^2 - y^2)$.

[3+2] [CO02] [BTL 3]

5) (i) Define Complete Integral, Particular Solution and Singular Solution of a Partial Differential Equation.

[3] [CO02] [BTL 1]

(ii) Solve the PDE: $pq + p + q = 0$, and find Complete, Singular and General Solution.

[2] [CO02] [BTL 5]

6) Classify the following partial differential equations as elliptic, parabolic, or hyperbolic:
(a) $u_{xx} + 4u_{yy} = 0$
(b) $u_{tt} = 9u_{xx}$
(c) $u_{xx} + 2u_{xy} + u_{yy} = 0.$

[5] [CO02] [BTL 2]

7) Let $S$ be the sample space when a pair of fair dice is tossed, i.e. $S = \{(a,b): a,b \in \{1,2,3,4,5,6\}\}, |S| = 36$. Define random variables: $X(a,b) = max(a,b)$, and $Y(a,b) = a + b$, then find the probability distribution of $X$ and $Y$, respectively. Also, find their expectation, variance, and standard deviation.

[10] [CO03] [BTL 3]

**Group B: Multiple Choice Questions.**

1) The PDE $\beta \frac{\partial^2 T(x,t)}{\partial x^2} + \frac{\partial T(x,t)}{\partial t} = 0$ is
(i) Parabolic
(ii) Hyperbolic
(iii) Elliptic
(iv) None of these

[1] [CO02] [BTL 1]

2) The PDE: $V_{xx} + V_{yy} + V_{zz} = 0$ is a
(i) Three-dimensional Heat Equation
(ii) Two-dimensional Wave equation
(iii) Three-dimensional Laplace Equation
(iv) Two-dimensional Heat Equation

[1] [CO02] [BTL 1]

3) A thin metal rod is insulated everywhere except at its ends. At $x = 0$and $x = l$, the ends are kept in ice. The initial temperature distribution is $T(x,0) = \sin(\pi x)$. Which of the following correctly represents the heat conduction problem for the rod?
(i) $T_{xx} + T_{tt} = 0;\ T(0,t) = T(l,t) = 0; T(x,0) = sin(\pi x)$



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(ii) $T_{xx} - T_{tt} = 0;\ T(0,t) = T(l,t) = 0; T(x,0) = sin(\pi x)$

(iii) $T_t = T_{xx}; T(0,t) = T(l,t) = 0; T(x,0) = \sin(\pi x)$

(iv) $T_{tt} = T_x; T(0,t) = T(l,t) = 0; T(x,0) = \sin(\pi x)$

[1] [CO02] [BTL 2]

4) If $z = f(x+y) + g(x-y)$, then $z$ satisfies which of the following PDEs?

(i) $z_{xx} - z_{yy} = 0$

(ii) $z_{xx} + z_{yy} = 0$

(iii) $z_{xx} - 2z_{xy} + z_{yy} = 0$

(iv) $z_{xy} = 0$

[1] [CO02] [BTL 2]

5) In the finite difference method, the second-order central difference approximation of $\frac{d^2y}{dx^2}$ at point $x_i$ is given by:

(i) $\frac{y_{i+1} - y_i}{h}$

(ii) $\frac{y_{i+1} - 2y_i + y_{i=1}}{h^2}$

(iii) $\frac{y_{i+1} - y_{i-1}}{2h}$

(iv) $\frac{y_{i+1} + y_{i-1} - 2y_{i-2}}{h^2}$

[1] [CO02] [BTL 1]

6) The independent random variables $X$ and $Y$ have variances 0.2 and 0.5 respectively. Let $Z = 5X - 2Y$. The variance of $Z$ is?

(i) 3

(ii) 4

(iii) 5

(iv) 7

[1] [CO03] [BTL 1]

7) The expected value of a discrete random variable $X$ with the following probability distribution is

| X | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| P(X) | 0.1 | 0.2 | 0.4 | 0.3 |

(i) 2.9

(ii) 3.0

(iii) 2.8

(iv) 3.1

[1] [CO03] [BTL 3]

8) For any random variables $X$ and $Y$, the covariance $\mathrm{Cov}(X,Y) = 0$ implies

(i) $X$ and $Y$ are independent

(ii) $X$ and $Y$ are dependent

(iii) $X$ and $Y$ are uncorrelated.

(iv) Both A and C.

[1] [CO03] [BTL 4]

9) If $f(x,y) = 3x + 2y$, then which of the following statements is true?

(i) $f(x,y)$ is nonlinear but convex



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(ii) $f(x,y)$ is both linear and convex
(iii) $f(x,y)$ is quadratic.
(iv) None of These

[1] [CO01] [BTL 2]

10) The objective function in a Linear Programming Problem (LPP) is generally of the form
(i) A nonlinear equation in several variables
(ii) A linear equation in one or more variables
(iii) A quadratic equation in one variable
(iv) A transcendental function

[1] [CO01] [BTL 1]

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**Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table**

| CO | Marks | BTL | Marks |
|---|---|---|---|
| CO01 | **17** | BTL 1 | **8** |
| CO02 | **20** | BTL 2 | **13** |
| CO03 | **13** | BTL 3 | **18** |
| | | BTL 4 | **6** |
| | | BTL 5 | **2** |
| | | BTL 6 | **3** |

**Bloom's Taxonomy Levels (attached for reference)**

Level 1 – Remember

Level 2 - Understand

Level 3 – Apply

Level 4 - Analyze

Level 5 - Evaluate

Level 6- Create



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